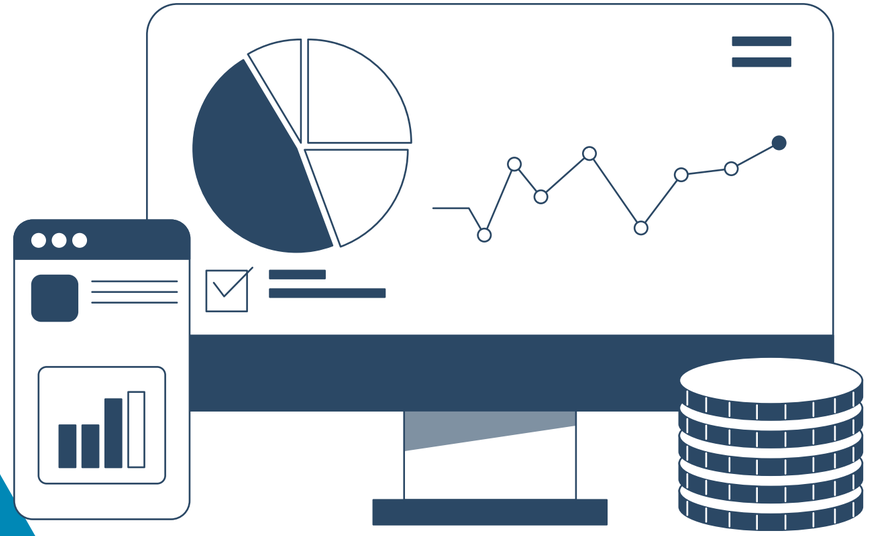
DATA ANALYTICS

30 days



DAY 25

Agenda

- Understand Feature Engineering
- Handle Missing Values professionally
- Merge Multiple Datasets
- Convert Categorical Data into Numbers
- Perform Group-wise Data Analysis

What is Feature Engineering?

Feature Engineering is the process of:

- Creating new features from existing data.
- Modifying existing features to improve their usefulness.
- Selecting the most relevant features for analysis or modeling.
- Transforming data into a format suitable for Machine Learning algorithms.

Data Preparation Pipeline

Raw Dataset

↓

Missing Value Handling

↓

Remove Duplicates

↓

Merge Datasets

↓

Encoding

↓

Feature Engineering

↓

Machine Learning Model

Managing Missing Values

Missing Value is a data value that is unavailable or not recorded in a dataset.

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
    "Name":["Rahim","Karim","Asha","John","Sara"],
    "Age":[22,np.nan,25,24,np.nan],
    "Salary":[50000,60000,np.nan,55000,52000],
    "Gender":["Male","Male","Female",np.nan,"Female"]
})
```

Name	Age	Salary	Gender
Rahim	22	50000	Male
Karim	NaN	60000	Male
Asha	25	NaN	Female
John	24	55000	NaN
Sara	NaN	52000	Female

```
df.isnull().sum()
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Salary"] = df["Salary"].fillna(df["Salary"].median())
df["Gender"] = df["Gender"].fillna(df["Gender"].mode()[0])
```

Encoding : Label Encoding

Encoding converts categorical (text) values into numerical values so Machine Learning algorithms can process them.

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df["Gender_Label"] =
le.fit_transform(df["Gender"])

df
```

Gender	Encoded
Male	1
Female	0
Male	1
Female	0

Encoding : Ordinal Encoding

Ordinal Encoding is the process of converting **ordinal categorical variables** into numerical values while **preserving their natural order or ranking**.

```
education = {
    "Bachelor":1,
    "Master":2,
    "PhD":3
}

df["Education_Ordinal"] =
df["Education"].map(education)

df
```

Education	Code
High School	1
Bachelor	2
Master	3
PhD	4

Encoding : One-Hot Encoding

One-Hot Encoding is the process of converting **nominal categorical variables** into multiple binary (0 or 1) columns, where each category is represented by a separate column.

```
pd.get_dummies(df, columns=["City"])
```

City	Dhaka	Khulna	Rajshahi
Dhaka	1	0	0
Khulna	0	1	0
Rajshahi	0	0	1

Merging Datasets

- Merging combines two or more datasets using a common column.
- Think of it like connecting two Excel sheets using a common ID.

```
pd.merge(customers, orders, on="Customer_ID", how="inner")
pd.merge(customers, orders, on="Customer_ID", how="left")
pd.merge(customers, orders, on="Customer_ID", how="right")
```

Customer_ID	Name
101	Rahim
102	Karim
103	Asha

Customer_ID	Product
101	Laptop
102	Mouse
104	Keyboard

Customer_ID	Name	Product
101	Rahim	Laptop
102	Karim	Mouse

Group-Based Analysis

Group-Based Analysis divides data into groups and performs calculations for each group independently.

```
sales = pd.DataFrame({
    "City":["Dhaka","Dhaka","Khulna","Khulna","Rajshahi"],
    "Category":["Laptop","Mobile","Laptop","Mobile","Laptop"],
    "Sales":[50000,30000,45000,25000,70000]
})
```

```
sales
```

City	Sales
Dhaka	500
Khulna	300
Dhaka	700
Khulna	400

```
sales.groupby("City")["Sales"].mean()
```

City	Average Sales
Dhaka	600
Khulna	350

Practice Assignment

```
data = pd.DataFrame({
    "Customer_ID":[101,102,103,104,105],
    "Name":["Rahim","Karim","Asha","John","Sara"],
    "City":["Dhaka","Khulna","Dhaka","Rajshahi","Khulna"],
    "Gender":["Male","Male","Female",np.nan,"Female"],
    "Age":[22,np.nan,25,24,np.nan],
    "Sales":[50000,30000,45000,25000,60000]
})
```

Practice Assignment

- Find missing values.
- Fill missing Age with the mean.
- Fill missing Gender with the mode.
- Apply Label Encoding to Gender.
- Group by City and calculate:
- Total Sales
- Average Sales
- Customer Count
- Create another DataFrame with customer membership information and merge it with the original data using Customer_ID.

THANK YOU

FOR YOUR ATTENTION